

# Hydrogel-Based Smart Patch for Early Detection of Post-Surgical Wound Infection Using pH Monitoring

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## Motivation

- Post-surgical wounds are susceptible to **Surgical Site Infections** after patients are discharged.
- Current monitoring relies on **periodic visual inspection during dressing changes**.
- A **hydrogel-based pH sensing patch** can provide **early warning of infection before visible symptoms appear**.



## State of the art

- Conventional wound assessment relies on **clinical observation of physical symptoms like redness, swelling**.
- Recent studies explore **smart dressings with sensors for pH, temperature, and moisture**.
- **Low-cost, continuous wound monitoring solutions are still needed**.



## Core technological idea

- Proposed device is a **hydrogel-based pH sensing patch**.
- A **hydrogel-based wound patch** absorbs wound exudate from the surgical site.
- Changes in wound pH are detected using a **pH sensor probe**.



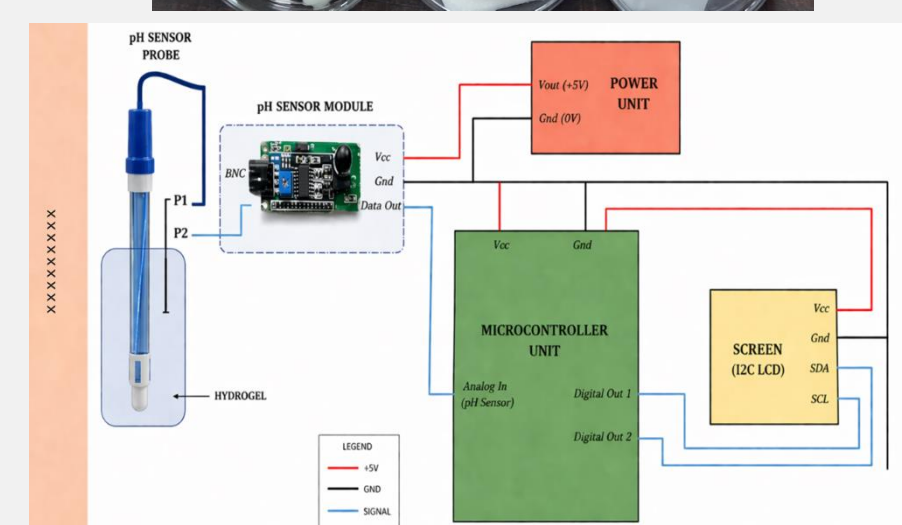
## Implementation & Achievements

- Developed a prototype integrating **hydrogel dressing, pH sensor probe, microcontroller and LCD display**.
- Fabricated and tested the **hydrogel sheet** for wound exudate absorption.
- Calibrated the pH sensor to differentiate **healing, chronic and infected wound conditions**.
- The device detects early biochemical signs of infection before pus formation.

|                        |          |
|------------------------|----------|
| Healing Surgical Wound | pH 5-6.5 |
| Chronic Wound          | pH 7-7.8 |
| Infected Wound         | pH 8-10  |



(Hydrogel Sheet)



(Circuit Diagram of Smart Wound pH Monitoring System)

## Reflection & Conclusion

- Smart biomaterials like hydrogels can serve as **interfaces between biological systems and electronic sensors**.

- Designing a sensing system that avoids **biofouling and signal drift** is difficult.
- Overall, If infection is not detected, it can spread to deeper tissues or bone, causing **Osteomyelitis**.

- Mostly useful for :
  - Orthopedic pin-site wounds
  - Diabetic-wound
  - Long-term suture
  - Post-surgical wound



(Ilizarov fixator)

## Outlook

- What could be done next
  - Integration of additional sensing parameters such as **temperature** or **moisture**.
  - Development of wireless health monitoring applications
  - Improve **post-operative wound care and remote patient monitoring**.

## References

- 1 Gethin, G. (2007). The significance of surface pH in chronic wounds. Wounds UK.
- 2 Schneider et al., Smart wound dressings for infection monitoring. Advanced Healthcare Materials, 2018.